

ParaLife Trees

Planting a Life in Paradise

Scotch Bonnet Pepper Seedling Grow Guide

Transplanting | Spacing | Feeding | Pest Control | Disease Control | Harvesting

Quick Fact	Recommended Range
Heat level	100,000-350,000 Scoville Heat Units (SHU)
Seedling transplant age	5-8 weeks from sowing, depending on seedling size and conditions
Best transplant size	12-15 cm / 5-6 inches tall, with strong roots and healthy leaves
For small farmers	Tray seedlings: harden off, transplant in evening, water immediately
For home gardens	2-6 plants can grow well in pots or beds with full sun and good drainage

WhatsApp: (876) 372-7415

This guide is designed for customers buying Scotch bonnet seedlings from ParaLife Trees - including small farmers planting trays and home gardeners planting a few plants.

1. Quick Rescue Chart: If You See This, Do This

Use this page first. Check under the leaves before you spray or fertilize. Most pepper pest problems begin on the underside of leaves and on soft new growth.

What you see	Most likely cause	First action today
Tiny white insects fly up when plant is touched	Whiteflies	Spray under leaves. Add yellow sticky traps. Repeat every 3-5 days until reduced.
Leaves curling and sticky; ants on plant	Aphids or whiteflies	Check under leaves. Spray mild soap solution. Control ants. Repeat every 3-5 days.
Fine webbing; dusty leaves; tiny speckles	Spider mites	Spray undersides. Do not let plant stay drought-stressed. Repeat every 3-5 days.
Bronze or silvery marks; distorted new leaves	Thrips or mites	Remove badly damaged growth. Treat early. Use sticky traps.
Holes in leaves or fruit; black droppings	Caterpillars/worms	Check early morning/evening. Handpick. Remove damaged fruit.
Yellow leaves plus wet soil	Overwatering/root stress	Reduce watering. Improve drainage. Raise pots off wet ground.
Wilting even though soil is wet	Root rot or bacterial wilt	Stop overwatering. Improve drainage. Remove badly collapsed plants.
Brown, black, or yellow spots on leaves	Leaf spot disease	Remove affected leaves. Avoid wetting leaves. Improve airflow.
Flowers falling off	Heat/water stress, pests, or too much nitrogen	Water consistently. Mulch. Reduce high-nitrogen feeding. Check flowers for pests.
Big green plant but few flowers	Too much nitrogen or low sunlight	Give 6-8 hours sun. Switch to flowering/fruiting support.

Important: If the plant is newly transplanted, do not rush with strong fertilizer. Let it settle for 7-14 days unless it is clearly nutrient-starved.

2. Common Pests: Fast Identification and Control

Pest	Signs	What to do
Whiteflies	Tiny white flying insects under leaves; yellowing; curling; sticky residue; black sooty mold can appear.	Spray under leaves every 3-5 days. Use yellow sticky traps. Remove badly infested leaves. Keep weeds down.
Aphids	Soft insects on new growth and under leaves; curled/wrinkled leaves; sticky honeydew; ants nearby.	Spray off with water. Use mild soap spray under leaves. Control ants. Repeat every 3-5 days.
Thrips	Bronze, dry, or silvery marks; inward curling; distorted new growth; flower damage.	Remove damaged leaves/flowers. Use sticky traps. Keep plants watered. Treat early.
Broad mites/spider mites	Tiny speckles, twisted new growth, webbing in spider mite cases; leaves may look dry or dusty.	Spray leaf undersides. Remove badly damaged leaves. Avoid drought stress. Repeat every 3-5 days.
Caterpillars/worms	Holes in leaves; chewed fruit; black droppings; damage can increase overnight.	Check plants early morning/evening. Handpick. Remove damaged fruit. Use approved biological control if available.
Crickets/cucumber beetles	Chewed seedlings, cut/damaged young plants, missing leaf pieces.	Keep field edges clean. Inspect new transplants daily for first 7-14 days. Protect weak seedlings.

Spray safety for customers: spray early morning or late evening, not in hot midday sun. Test any spray on a few leaves first and check the plant the next day.

3. Common Diseases and Sanitation Warnings

Disease/problem	Signs	What to do
Viral disease / mosaic	Mosaic or mottled light/dark green leaves, severe curling, stunting, deformed leaves or fruit.	Control whiteflies/aphids/thrips early. Remove badly infected plants. Wash hands and tools. Do not compost infected plants.
Tobacco mosaic virus risk	Mosaic patterns, leaf distortion, weak growth. Virus can move by contaminated hands/tools.	Wash hands with soap before handling peppers, especially after smoking or handling tobacco products. Do not smoke while handling seedlings.
Bacterial spot / leaf spot	Brown, black, or yellow leaf spots. Leaves may yellow and drop, worse in wet conditions.	Remove affected leaves. Avoid wetting leaves. Improve spacing and airflow. Disinfect tools.
Fusarium/bacterial wilt	Plant wilts suddenly; may not recover after watering; stems/roots may show internal browning.	Remove badly affected plants. Avoid replanting peppers in the same infected spot. Improve sanitation and drainage.
Southern blight / stem rot	White fungal growth near soil line; plant collapses; stem base rots.	Remove plant and infected soil near stem. Keep mulch away from stem. Improve airflow and drainage.
Anthraxnose fruit rot	Sunken spots on fruit; fruit may rot before harvest.	Remove infected fruits. Avoid overhead watering. Harvest regularly.

Tobacco/smoker warning: Pepper plants can be affected by tobacco mosaic-type viruses. Smokers and workers who handle tobacco products should wash hands with soap and water before touching seedlings, leaves, trays, pruning tools, or fruit. Use clean gloves for tray handling and transplanting.

4. Transplanting Seedlings From Trays

This is the most important section for customers buying seedlings. A healthy seedling can still suffer if it is moved from tray to field or pot too quickly, planted too deep, or allowed to dry out during transplanting.

Seedling readiness numbers

Item	Target range
Age from sowing	5-8 weeks, depending on growing conditions
Height	12-15 cm / 5-6 inches is a good transplant size
Leaves	At least 4-6 true leaves; leaves should be green and not badly curled
Roots	Root ball should hold together, but should not be severely root-bound
Transplant time	Evening is best; early morning is second best
First week after transplant	Keep well-watered for 7 days while roots settle

Step-by-step transplanting

- Water the tray 1-2 hours before removing seedlings.
- Handle seedlings by the root ball or leaves, not by squeezing the stem.
- Keep the root ball intact. Bare-root or damaged-root seedlings recover more slowly.
- Plant at the same depth the seedling was growing in the tray or pot.
- Press soil gently around the root ball so there are no large air pockets.
- Water immediately after transplanting.
- Check daily for the first 7 days. Replace dead or weak plants early if planting for production.

Do not let tray seedlings sit in direct hot sun after purchase while waiting to plant. Keep them watered, bright, and protected until transplanting.

5. Hardening Off Before Full Sun

Hardening off means training seedlings to handle stronger sun, wind, and field conditions before they are planted permanently.

7-day hardening plan

Day	Sun exposure	Care note
Day 1-2	2-3 hours morning sun	Keep seedlings watered; avoid midday sun.
Day 3-4	3-5 hours sun	Allow more airflow; do not let trays dry out.
Day 5-6	5-7 hours sun	Expose to near-final conditions; check for wilting.
Day 7	6-8 hours sun if plants stay firm	Ready to transplant in evening if roots and leaves look healthy.

If seedlings wilt badly during hardening, reduce sun exposure for 24 hours, water properly, and restart with fewer hours of sun. Hardening should toughen the plant, not burn it.

For small farmers planting trays: schedule hardening so seedlings are field-ready before labour arrives. Avoid buying/collecting seedlings and leaving them stressed in trays for several extra days.

6. Spacing for Small Farmers and Home Gardens

Use wider spacing for easier harvesting, better airflow, and lower disease pressure. Use closer spacing only when you can manage pruning, airflow, irrigation, feeding, and pest checks properly.

Planting situation	Spacing	Approximate plant count
Beginning small farmer field layout	3 ft between plants x 5 ft between rows	About 3,630 plants per acre
Medium intensive field layout	60 cm between plants x 76 cm between rows	About 21,527 plants per hectare
High intensive field layout	60 cm x 60 cm	About 26,898 plants per hectare
Home garden minimum	18 inches between plants	Use only with good airflow and pest checks
Better home garden spacing	24 inches between plants	Easier care and better airflow
Containers	Minimum 3 gallons; better 5 gallons or larger	1 plant per pot

For customers buying 2-6 plants: use 5-gallon pots if possible, or plant 2 ft apart in a bed. Make sure each plant gets 6-8 hours of sun.

7. Soil, Sun, Climate, and Water

Need	Target range	Problem if outside range
Sunlight	6-8 hours direct sun daily	Less than 5 hours often gives weak growth and fewer flowers.
Soil pH	6.0-7.0 acceptable; 6.5 is a strong target	Low or high pH can reduce nutrient uptake.
Soil temperature at transplant	Above 60 F / 15.5 C	Cold soil slows new roots and recovery.
Soil type	Loose, well-drained soil with compost	Heavy waterlogged clay increases root disease.
Water after transplant	Water immediately; keep well-watered for first 7 days	Dry root ball causes transplant shock and wilting.
Critical water periods	First week after transplant, flowering, fruit set, fruit development	Water stress can cause flower drop and small fruit.
Mulch	Keep mulch 2-3 inches away from stem	Mulch touching stem can increase stem rot.

Watering rule for pots: check daily in hot weather. Water when the top 1 inch of mix begins to dry, but do not leave the pot sitting in water.

8. Feeding Schedule

Best feeding depends on soil test, potting mix, rainfall, and plant load. These ranges give customers a confident starting point without encouraging overfeeding.

Stage	Timing	Feeding direction
At transplanting	Day 0	Compost or well-prepared soil. For field systems, use a complete fertilizer program based on soil test where possible.
After settling	10-14 days after transplant	Light balanced feed if plant is actively growing. Avoid strong feed on wilted/stressed plants.
Flowering begins	Around 39-44 days after transplant	Reduce heavy nitrogen. Support flowering with balanced nutrition and steady moisture.
Fruiting support	25-30 days after flowering begins	Support potassium and calcium; avoid drought stress.
During harvest	Every 4-6 weeks in ground; every 2-4 weeks in pots	Light maintenance feeding based on plant color, fruit load, and soil quality.

Overfeeding warning: too much nitrogen can give a large green plant with fewer flowers, more soft growth, and more pest pressure.

For commercial-scale customers, use soil and leaf tissue analysis when possible. Fertilizer rates should be adjusted to soil fertility, plant population, irrigation, rainfall, and expected yield.

9. Growth Timeline With Numbered Ranges

Stage	Estimated time range
Seed germination	7-21 days; some guides report 7-12 days under strong conditions
Seedling ready for transplant	5-8 weeks from sowing
Hardening off	7 days before transplanting
Root settling after transplant	7-14 days
Strong new growth after transplant	14-28 days
First flowers	About 39-44 days after transplant
Early fruit set	Usually begins after successful flowering/pollination; monitor from 45-70 days after transplant
First harvest window	About 80-120 days after transplant, depending on variety, plant size at purchase, and growing conditions
Harvest frequency once producing	Every 5-10 days

For customers buying established seedlings, the countdown starts from transplanting day. A larger, older seedling may flower sooner than a younger seedling, but only if it is not stressed during transplanting.

10. Healthy Leaf Color and Problem Signs

Leaf appearance	Likely meaning	Action
Medium to deep green	Generally healthy	Keep watering and pest checks consistent.
New leaves light green	Normal if they darken within 7-14 days	Watch; do not overfeed immediately.
Pale green all over	Low nitrogen, poor soil, low light, or water stress	Check sun and water first; add compost/light feed if needed.
Yellow lower leaves	Age, overwatering, low nitrogen, or root stress	Check drainage and soil moisture. Remove badly yellow leaves.
Brown leaf edges	Heat, fertilizer burn, underwatering, or salt buildup in pots	Water deeply; avoid strong fertilizer; flush pots if needed.
Curling leaves	Whiteflies, aphids, mites, thrips, heat, or virus	Inspect under leaves. Treat pests early. Remove severely infected plants.
Mosaic/mottled pattern	Possible viral disease	Wash hands/tools. Remove badly affected plants. Control insect vectors.

11. Flowering and Fruiting Problems

Problem	Likely cause	What to do
Flowers drop before fruit forms	Heat, water stress, pests, poor pollination, or too much nitrogen	Water consistently, mulch, check flowers for thrips, reduce nitrogen.
Plant has leaves but no flowers	Too much nitrogen or less than 6 hours sun	Give 6-8 hours sun; switch to fruiting support.
Small peppers	Water stress, weak roots, poor feeding, or pest damage	Keep moisture steady, feed lightly, inspect leaves and flowers.
Deformed peppers	Thrips, mites, virus, or poor pollination	Check new growth and flowers; remove badly affected fruit.
Fruit rots before harvest	Anthrachnose, wet foliage, pest injury, or poor airflow	Remove infected fruit, avoid overhead watering, improve spacing.

Pollination tip for home gardeners: peppers are self-pollinating, but gentle plant movement, breeze, and pollinator visits can improve fruit set. Avoid spraying open flowers when bees are active.

12. Harvesting Guide

Harvest item	Specific guidance
Green harvest	Harvest when fruit reaches full size and is firm.
Ripe harvest	Harvest when fruit reaches mature variety color: yellow, orange, red, chocolate/brown, or other type.
Harvest frequency	Every 5-10 days during active production.
Handling	Use scissors/pruner to avoid breaking branches. Wear gloves if harvesting many peppers.
After harvest	Keep fruit shaded and cool. Sort damaged or diseased fruit away from clean fruit.

Harvesting regularly encourages continued production and helps remove damaged fruit before pests and diseases spread.

13. Weekly Care Checklist

Frequency	Checklist
Every 1-2 days	Check soil moisture; look under leaves; check for whiteflies; check for wilting; inspect flowers and young fruit.
Every 7 days	Remove yellow/diseased leaves; weed around plants; check for ants, aphids, whiteflies, mites, thrips, caterpillars; check stakes.
Every 2-4 weeks	Feed lightly if plant is growing and fruiting; top up compost or mulch; inspect airflow and crowding.
Before handling trays	Wash hands; clean tools; avoid tobacco handling before touching seedlings.
After handling diseased plants	Wash hands, disinfect tools, and handle healthy plants separately.

Photo support checklist

If you need help, send ParaLife Trees clear photos of: 1) the full plant, 2) top leaves, 3) underside of leaves, 4) stem base and soil/pot, 5) any insects, spots, damaged flowers, or damaged fruit.

14. ParaLife Trees Support and Source Notes

Need help with your Scotch bonnet plant?

Send clear photos by WhatsApp and describe when the problem started, when you transplanted, how often you water, whether it is in a pot or the ground, and how much sun it receives daily.

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This guide is a practical customer guide, not a pesticide label or professional diagnosis. Always follow product labels and local agricultural guidance when applying fertilizers, pesticides, fungicides, or biological controls.

Source notes used to prepare this guide

No.	Source note
1	Ministry of Agriculture Jamaica / FAO, Growing Scotch Bonnet Pepper in Jamaica. Used for seedling transplant size, hardening, transplanting steps, irrigation, flowering timing, fertilizer examples, pests and diseases.
2	CARDI, Caribbean Hot Pepper Production and Post Harvest Manual. Used for pest/disease descriptions, spacing intensity, fertility and moisture principles.
3	Delaware State University Cooperative Extension, Scotch Bonnet. Used for small-farmer spacing, soil pH, 6-8 week transplant window, and tray/field establishment notes.
4	Missouri Botanical Garden, Bayer Vegetables, UMass Extension and Michigan State University Extension guidance on tobacco mosaic-type virus sanitation. Used for the tobacco/smoker handwashing warning.